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Globalization of wild capture and farmed aquatic foods

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Aquatic foods are highly traded, with nearly 60 million tonnes exported in 2020, representing 11% of global agriculture trade by value. Despite the vast scale, basic characteristics of aquatic food trade, including species, origin, and farmed vs wild sourcing, are largely unknown due to the reporting of trade data. Consequently, we have a coarse picture of aquatic food trade and consumption patterns. Here, we present results from a database on species trade that aligns production, conversion factors, and trade to compute apparent consumption for all farmed and wild aquatic foods from 1996 to 2020. Over this period, aquatic foods became increasingly globalized, with the share of production exported increasing by 40%. Importantly, trends differ across aquatic food sectors. Global consumption also increased by 19.4% despite declining marine capture consumption, and some regions became increasingly reliant on foreign-sourced aquatic foods. To identify sustainable diet opportunities among aquatic foods, our findings, and underlying database enable a greater understanding of the role of trade in rapidly evolving aquatic food systems.

Aquatic food systems are an important source of human nutrition[1], livelihoods[2], and revenue[3] throughout the world. Aquatic foods also show promise to reduce environmental pressures of food production due to low average resource use and emissions[4]. However, aquatic foods are incredibly diverse, comprising over 2400 marine and freshwater species that are captured and farmed with a range of methods[5]. Consequently, aquatic foods vary widely in their nutrient composition[1] and associated environmental pressures[4]. This has prompted work to identify and support aquatic food systems that improve nutrition, sustainability, and human well-being[6–9]. With the international trade of aquatic food representing 40% of production[3], accounting for trade is central to meeting these objectives.

Trade brings a range of benefits and risks for food security, resilience, and sustainability. Benefits include providing consumers with diverse and out-of-season foods, supplying products at lower prices, stimulating local economic growth, diversifying sourcing in the face of local shocks, and reducing environmental impacts when products are sourced from regions better suited for production[10]. However, risks stemming from globalization include accelerating the nutrition transition to unhealthy diets[11], undermining domestic production by suppressing prices[12], exposing local markets to international shocks[13], degrading local environments to meet distant market demand[14,15], and facilitating shifting production to locations with relaxed environmental and labor regulations[16,17].

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Which trade-related benefits and risks are experienced, and by who is context-dependent. Unfortunately, our understanding of the distribution of global benefits and risks is limited by the low species resolution of global trade data relative to the vast diversity of aquatic foods, and because internationally reported trade data does not indicate the country of harvest. Consequently, we only have a coarse picture of the basic features of the aquatic food trade, including the geographical origin and production method (wild or farmed)[18,19]. Coarse trade data further places profound constraints on understanding aquatic food consumption patterns and therefore the potential role of aquatic foods in sustainable and resilient food systems.

Limited resolution of aquatic food consumption arises from a fundamental mismatch between production and trade data: production from capture fisheries and aquaculture is reported as species or species groups (e.g., *Salmo salar* or *Oncorhynchus* spp.) in terms of live weights whereas trade is reported as commodities (e.g., canned salmon) in terms of product weight, generally without farmed vs wild designations. Converting commodity trade to species trade is difficult because one species can contribute to multiple commodities (e.g., *S. salar* can be converted into whole frozen salmon or salmon filets), a single commodity can be made up of multiple species (e.g., salmon filets can be made from *S. salar* or *Oncorhynchus tshawytscha*), and a traded commodity can be converted through processing and exported again (e.g. whole frozen salmon processed into salmon filets).

To improve understanding of global aquatic food trade and the associated implications for food security, resilience, and sustainability, we present a global database of species trade flows for all farmed and wild aquatic foods from across marine and inland waters from 1996 to 2020. The Aquatic Resource Trade in Species (ARTIS) database consists of over 2400 species/species groups, 193 countries, and over 35 million bilateral records. We estimated species trade flows by modeling each country's conversion of wild and farmed production into commodities, conversion of imported commodities through processing, and apparent consumption. We then connected estimated species mixes and processing of foreign-sourced products to bilateral trade data to disaggregate global flows of aquatic foods. ARTIS improves upon previous efforts by estimating annual species-level trade across production methods and habitats rather than providing an aggregate snapshot of capture and aquaculture trade[20], and by accounting for processing losses and foreign processing[21]. The resulting data and code accompanying this paper will serve as a critical resource for future research on aquatic food trade and consumption.

Using ARTIS, we characterize the global farmed and wild aquatic food trade, including all fish and aquatic invertebrate species destined for human consumption. We first detail the evolution of trade in marine and inland capture and aquaculture products, providing new measures of the degree of globalization across aquatic foods. We show that aquatic food trade more than doubled from 1996 to 2019, with aquaculture growing at a faster rate, but capture fisheries are still responsible for the majority of the aquatic food trade. We also demonstrate large differences in the export orientation of different aquatic food sectors, with a third of all marine production destined for export but only around 6% of inland aquaculture production being exported. Second, we evaluate how bilateral flows of aquatic foods have shifted since 1996, demonstrating shifts in key importers and exporters of aquaculture and marine capture products. Finally, we present trends in aquatic food apparent consumption, including shifts in import dependence. We find that although aquatic food consumption has increased overall, aquaculture is responsible for this trend, as consumption of capture fishery products has declined. However, both the production method and geographical sourcing patterns vary greatly by region, with the largest increases in aquaculture consumption and domestic sourcing in Asia, and growing aquaculture consumption but increasing import reliance in Europe and North America. Across each of these areas, we contextualize our findings with the implications for food security, sustainability, and resilience.

Results and discussion

Trends in aquatic food globalization

Globalization describes the degree of international connectedness, which can be characterized by increasing flows of input, intermediate, and final products among countries. Globalization exposes countries to external shocks, while also serving as a buffer against local shocks. Recent work on trade characteristics associated with systemic risk to shocks suggests higher exposure when networks are densely connected and concentrated, and when countries are highly dependent on imports[22–24]. By disaggregating global aquatic food trade, we can evaluate the structural features of the aquatic food trade associated with resilience to shocks.

Aquatic food exports more than doubled from 1996 to 2019 (28.1–59.2 mil t; Fig. 1a), on par with cereal products[25]. Over that period, both farmed and wild exports increased, though aquaculture grew faster, more than tripling, whereas capture exports grew by 74%. Corresponding with the start of the COVID-19 pandemic, global aquatic food exports declined 4% in 2020 relative to 2019, with a 3% decline in capture exports, but a 0.8% increase in aquaculture exports, highlighting differential impacts by sector (Fig. 1a). Despite aquaculture comprising half of aquatic food production, capture fishery products still constitute 60% of exports.

Another measure of the degree of globalization is the share of production exported. Domestic exports increased from 15.7% to 21.5% of production between 1996 and 2019, while total exports, which include export of foreign-sourced products, reached 33.5% of all production in 2019—slightly lower than the FAO estimates 36% for all aquatic resources[3]. Trade statistics do not generally distinguish between exports sourced from domestic production and those sourced from imported products, including products moving through intermediate countries or undergoing foreign processing. Consequently, total exports represent a higher percentage of production. For comparison, the share of cereal production exported grew from around 10% in the late 1990s to 17% in the 2020s[26]. Increasing marine capture exports despite stagnating production resulted in marine capture products having the greatest share of production destined for export (32.5% of production) and the largest increases in the share exported (Fig. 1b). Aquaculture production more than doubled from 1996 to 2019, but aquaculture exports grew even faster, increasing the share exported (Fig. 1b). Despite increases, inland aquaculture still had the lowest share of production destined for export in 2019 (domestic exports represented only 6.3% of production) (Fig. 1b). This finding clarifies standing debates about the orientation of aquaculture and export trends suggest a need to consider international markets when crafting nutrition-sensitive policies[27–29]. Understanding the retention and foreign flow of aquatic foods and their associated nutrients is central to current work on equity and justice within aquatic food systems. Consequently, this information is key to monitoring the progress of nutrition-sensitive policies and for crafting policies that appropriately reflect the global nature of aquatic foods.

We also identified shifts in the structure of the aquatic food trade networks. We found that aquatic food trade became more connected with the average number of export partners nearly doubling from 1996 to 2019 (from 21.9 to 41.5; Fig. 1c). Marine capture networks are most highly connected, followed by marine aquaculture, with inland capture and aquaculture trade being the least connected.

Since 1996, aquatic food exports have become moderately less concentrated, with only 18 countries comprising 75% of exports in 1996 vs 21 countries in 2019. Compare this with crops where just 7 countries and the EU account for 90% of wheat exports and just four countries account for >80% of maize exports[26]. Declining

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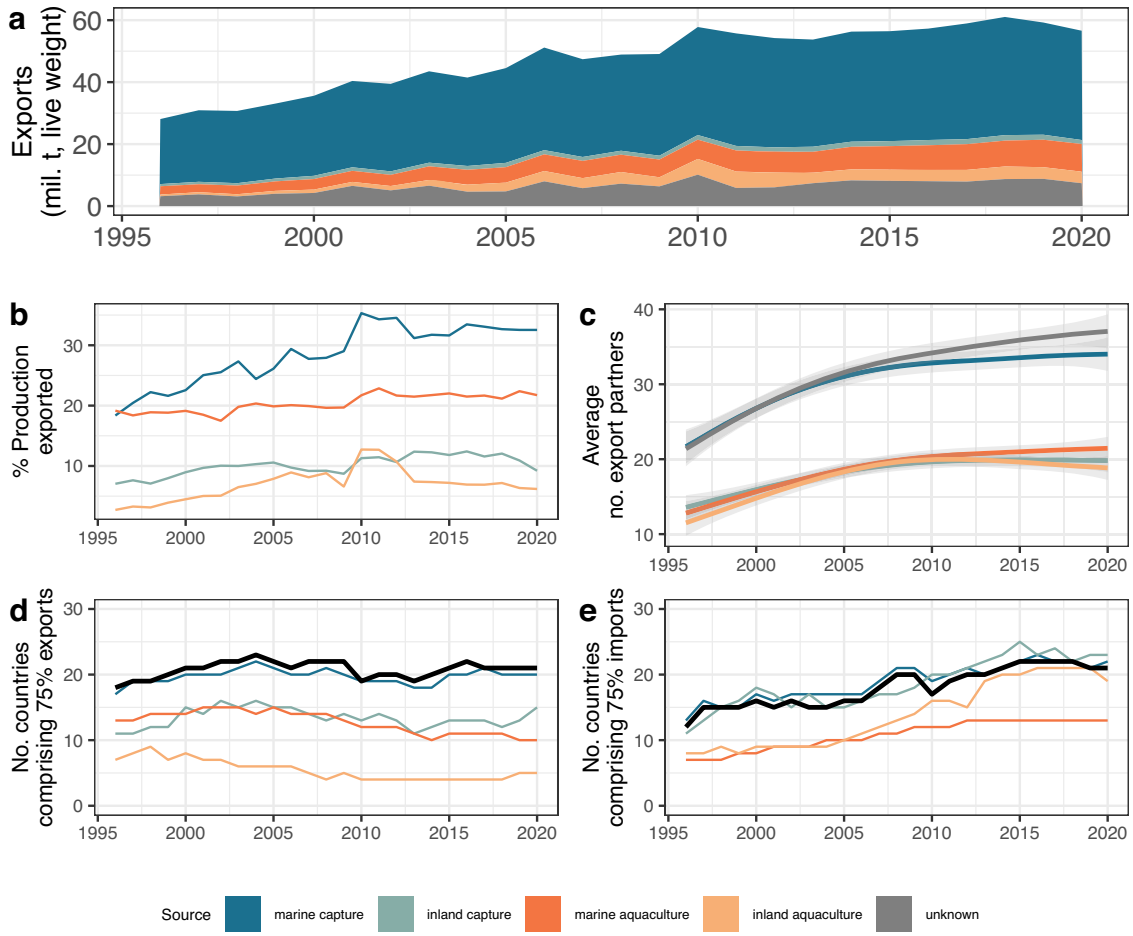


Fig. 1 | Increases in global export of aquatic foods. a Exports of global marine and freshwater aquaculture and fishery products (in million t, live weight equivalent) from 1996 to 2020. **b** Percent of global marine and freshwater aquaculture and fishery production exported, excluding re-exports. **c** Average number of export partners (out-degree) by production method and environment, with the gray bands representing the 95% confidence interval. **d** Number of countries comprising 75% of global exports by production method and environment with the global total in the black line. **e** Number of countries comprising 75% of global imports by production method and environment with the global total in the black line.

concentration is driven by capture fishery exports, whereas aquaculture exports became somewhat more concentrated (Fig. 1d). Aquaculture export concentration corresponds to the high concentration of aquaculture production in a few regions. Similarly, the concentration of trade for individual species tends to be much higher. Divergent trends in trade features and differences among aquatic food groups suggest differences in the degree and types of trade shock risks across aquatic foods, as was observed in the responses to COVID-19[30]. Understanding the risk of shocks across foods is a priority research area, as trade-related risks and aquatic food systems are underrepresented in the food systems shock literature[13].

Though aquatic food production, distribution, and consumption remain highly uneven[31], we found declining import concentration, with 12 countries comprising 75% of imports in 1996 vs 21 countries in 2019 (Fig. 1e). More dispersed import patterns are likely associated with growing populations and expanding middle classes and urbanization, particularly in low- and middle-income countries, which often drive increasing aquatic food demand[3,32]. Yet, the relationship between aquatic food demand and income varies across aquatic foods, with demand generally increasing with income for higher quality fish but falling for lower quality fish[32].

Shifts in global flows of aquatic foods

Given the geographic patchiness of capture and aquaculture production, trade helps meet aquatic food demand in many countries. Aquatic food imports are especially important where per capita demand is rising, aquaculture is limited, wild fishery catch is stagnant, and aquatic foods play an important nutritional role. Corresponding to the geographical variability, we find the top importers, exporters, and bilateral flows to differ by habitat and farmed vs wild source, underscoring the importance of disaggregating trade (Supplementary Figs. 1–3). For example, Asia and Europe, and to a lesser extent, North America recently dominated marine capture and aquaculture trade networks whereas Asia dominates all inland aquatic food trade (Supplementary Fig. S1).

At the country level, although some countries rank among the top traders across all production methods, such as China for exports and China, the United States and Japan for imports, many countries are only top traders for one (Supplementary Fig. 2). Between 1996 and 2020, China and Russia took over as the top two marine capture exporters, with approximately half of China's marine capture exports representing re-exported products (Supplementary Fig. 2), which aligns with previous findings that 75% of China's total aquatic food imports are re-exported[33]. China, the United States, and Japan are the top importers of marine capture products (Supplementary Fig. 3). Meanwhile, Norway and Chile rank highest in marine aquaculture exports, with the United States and Japan leading imports (Supplementary Figs. 2–3). Inland aquatic food trade is dominated by aquaculture, with the highest exports from Vietnam and China and the highest imports by the United States, Japan, and South Korea

(Supplementary Figs. 2–3). In general, inland production is oriented more toward domestic consumption, and what is exported tends to stay within the region, particularly within Asia (Fig. 2 and Supplementary Fig. 1).

The patterns of top importers are reflected in the patterns of top consumption of foreign aquatic foods by habitat and production group, though the patterns of top flows are more distinct (Fig. 2). For example, although the United States is among the top importers for all four habitat and production groups, the sourcing varies, with the largest marine capture sourcing from Canada and Russia, the largest marine aquaculture sourcing from Chile, and its largest inland aquaculture sourcing from China and Vietnam (Fig. 2). The major sourcing flows are closely related to the production geographies of the top traded species by production group (Fig. 2). There is very little production method overlap in the top consumed traded species and the distributions are relatively flat, apart from marine aquaculture, which is dominated by white leg shrimp (*Litopenaeus vannamei*) and Atlantic salmon (*S. salar*) (Fig. 2).

Intraregional trade is generally higher than interregional trade due to shorter transport distances, historical ties, patterns of aquatic food preferences, and established regional trade agreements[34]. We find this pattern largely holds for aquatic food trade as intraregional trade is the highest for Asia, Africa, and Europe (Fig. 3b). However, Oceania and North and South America all have the largest export to Asia, rather than to other countries in the same region (Fig. 3b). Since 1996, trade increased or remained approximately stable between nearly all regional trade pairs, other than within North America (Supplementary Fig. 4). At the country level, trade increased between two-thirds of trade pairs. Despite trade increasing with partners across the globe, trade within Asia, Europe, and Africa grew faster. The largest average annual growth increases occurred for trade within Asia and Europe, followed by trade between Europe and Asia (Fig. 3b and Supplementary Fig. 4). Our trade estimates are ultimately from reported trade and therefore do not capture informal and unreported trade networks. Though estimated unreported trade is not globally available, it can be significant, especially for neighboring countries. For example, informal exports from Benin to Nigeria are estimated to be more than five times the formal exports[35]. Including informal trade would therefore likely strengthen some intraregional trade patterns.

Increasing global trade, along with distant water fishing, drives an expanding divide between aquatic food production and consumption[36]. Distancing consumers from the environmental impacts stemming from production has been proposed to reduce incentives for improving environmental standards and weaken price signals related to fishery sustainability[37]. However, the specific impacts resulting from trade will depend upon the environmental management context. Complex international supply chains also pose a challenge for traceability, raising sustainability concerns, including the risk of mislabeled[38] and illegally sourced[39] products entering markets. We find increasing volumes of products moving through intermediate countries, either in transit or imported for processing and re-exported, which poses a traceability challenge (Supplementary Fig. 5). Certification and import monitoring schemes represent two tools aimed at improving traceability, and ultimately, sustainable sourcing. However, evidence of the effectiveness of aquatic food supply chain transparency initiatives is mixed[40]. Our findings on increasing globalization across the aquatic food sector underscore the importance of evaluating the effectiveness and social impacts of these sustainability tools across a range of settings, while the ARTIS database enables future work on this topic.

Across regions, Europe and North America have the highest net imports while South America has the highest net exports (Fig. 3a). Least developed countries collectively are net exporters of aquatic foods across all production methods, with net exports more than tripling between 1996 and their 2018 peak (Supplementary Fig. 6). Least developed country net exports are dominated by marine capture products and transfer of aquatic foods from least developed countries are likely even higher when catch by foreign fleets are considered, which represents a related but distinct mechanism by which aquatic resource are removed from countries' waters. Net exports of aquatic foods may be economically beneficial to least developed countries where high-value species are exported and revenue is used to purchase other foods[41]. However, economic and political barriers inhibit wealth-based benefits from being realized[31,42]. Further, recent work exploring the movement of nutrients derived from fisheries suggests international trade is driving the redistribution of essential micronutrients from areas of high deficiency in middle- and low-income countries to developed nations with greater nutrient security[43].

Aquatic food consumption

Since direct measurements of human food consumption (e.g., dietary intake) are not collected globally, it is often represented by apparent consumption. Apparent consumption is calculated as production plus imports minus exports and waste. Trade is therefore central to estimating consumption and has historically limited understanding of aquatic food consumption patterns. By estimating species-level trade, we estimate the apparent consumption of aquatic foods by species/species group, production method, and geographical origin.

Globally, annual aquatic food apparent consumption increased from 15.7 kg/capita in 1996 to 20 kg/capita in 2019 (Fig. 4a). Our estimates are similar to FAOSTAT[25], which reports global aquatic food consumption at 15.6 kg/capita/year in 1996 and 20.7 kg/capita/year in 2019. Given the similarity in consumption, this finding reaffirms the nutritional importance of aquatic foods, which supply 17% of animal protein globally, and over 20% of animal protein for over 3.3. billion people[44]. We found aquatic food consumption increased across all regions outside of North America, which was relatively stable, and South America, where aquatic food consumption declined 35.5% (Fig. 4b). Global increases were driven by inland and marine aquaculture, which increased by 162% (from 2.54 kg/capita/year in 1996 to 6.64 kg/capita/year in 2019) and 177% (from 1.23 kg/capita/year in 1996 to 3.4 kg/capita/year in 2019), respectively. Meanwhile, inland capture consumption grew from 1.16 kg/capita/year in 1996 to 1.51 kg/capita/year in 2019, while marine capture consumption declined 21.5% (from 10.8 kg/capita/year in 1996 to 8.45 kg/capita/year in 2019). Nevertheless, capture still makes up 49% of global aquatic food consumption, with its contribution to regional aquatic food consumption ranging from 76% in Oceania to 38% in Asia, where farmed consumption overtook wild consumption in 2008.

Estimating the foreign versus domestic source of consumption requires identifying the share of production retained in the country and tracking products that undergo foreign processing but are imported again. By estimating the source of traded aquatic foods, we can therefore track changes in reliance on foreign-sourced products. Globally the share of foreign-sourced consumption increased modestly, from 20.1% in 1996 to 20.9% in 2019 (Fig. 4c). However, patterns vary greatly across regions with countries in Asia and South America dominated by domestic supply (11% and 20% foreign in 2019, respectively), but countries in Europe dominated by foreign supply (72% foreign in 2019) in 2019 (Fig. 4d). High reliance on foreign-sourced foods can pose food security risk[45,46], though it is not clear the extent to which these risks exist across aquatic foods. Nevertheless, countries have enacted policies to protect domestic supplies, including developing food stocks and subsidizing domestic food production[47]. The United States has used foreign dependence on aquatic foods as motivation for a suite of policy changes to boost domestic production, including expanding aquaculture and opening marine protected areas to fishing[18].

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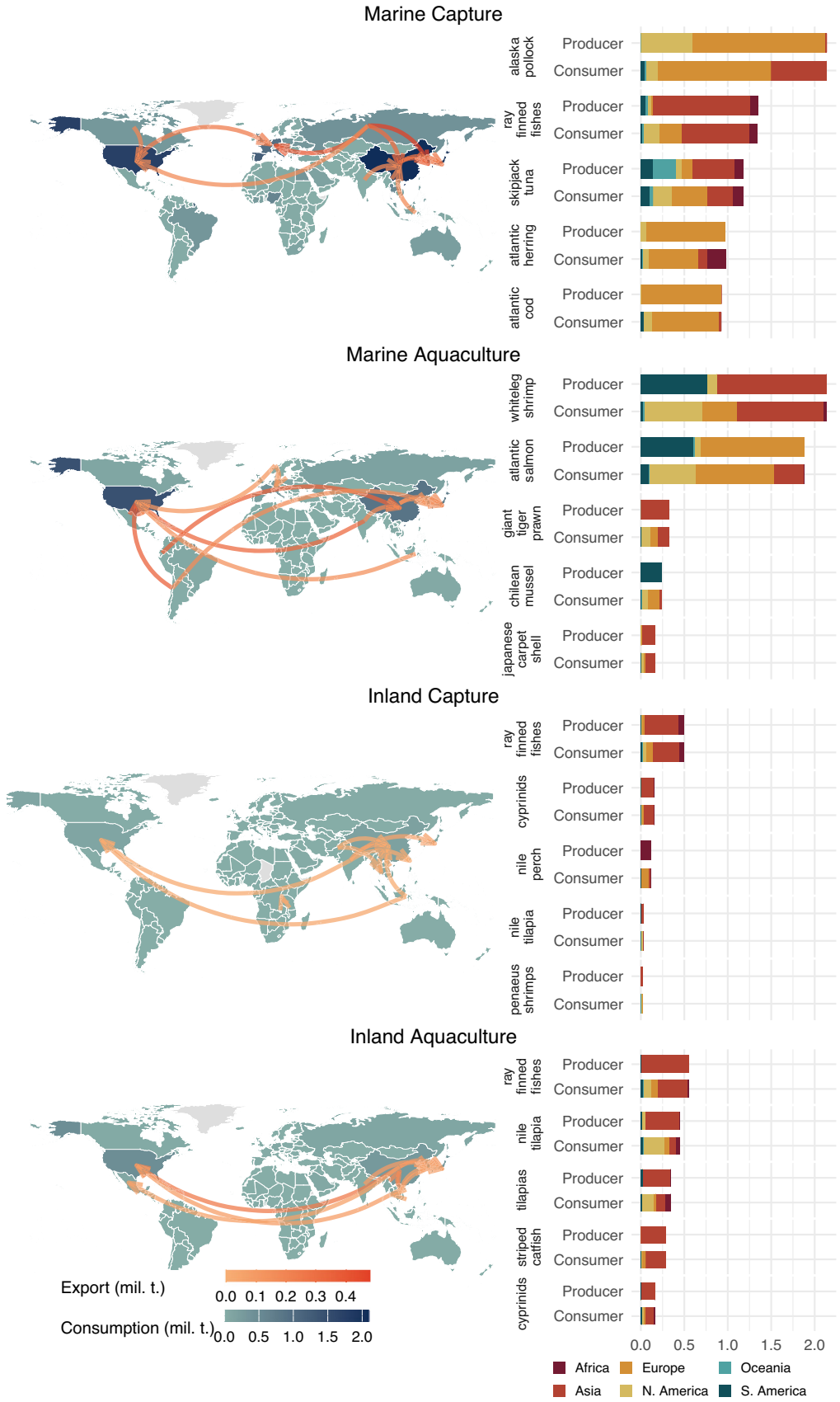


Fig. 2 | Top trade flows and traded species by production source (habitat and method). Maps of national aquatic food consumption (in million tonnes, live weight equivalent) represented by fill color for marine capture, marine aquaculture, inland capture, and inland aquaculture, along with the top ten trade flows, with the arrow color indicating export volume. Accompanying each map is a ranked bar chart of the top five traded scientific names, each with two stacked bars to indicate the producing and consuming regions.

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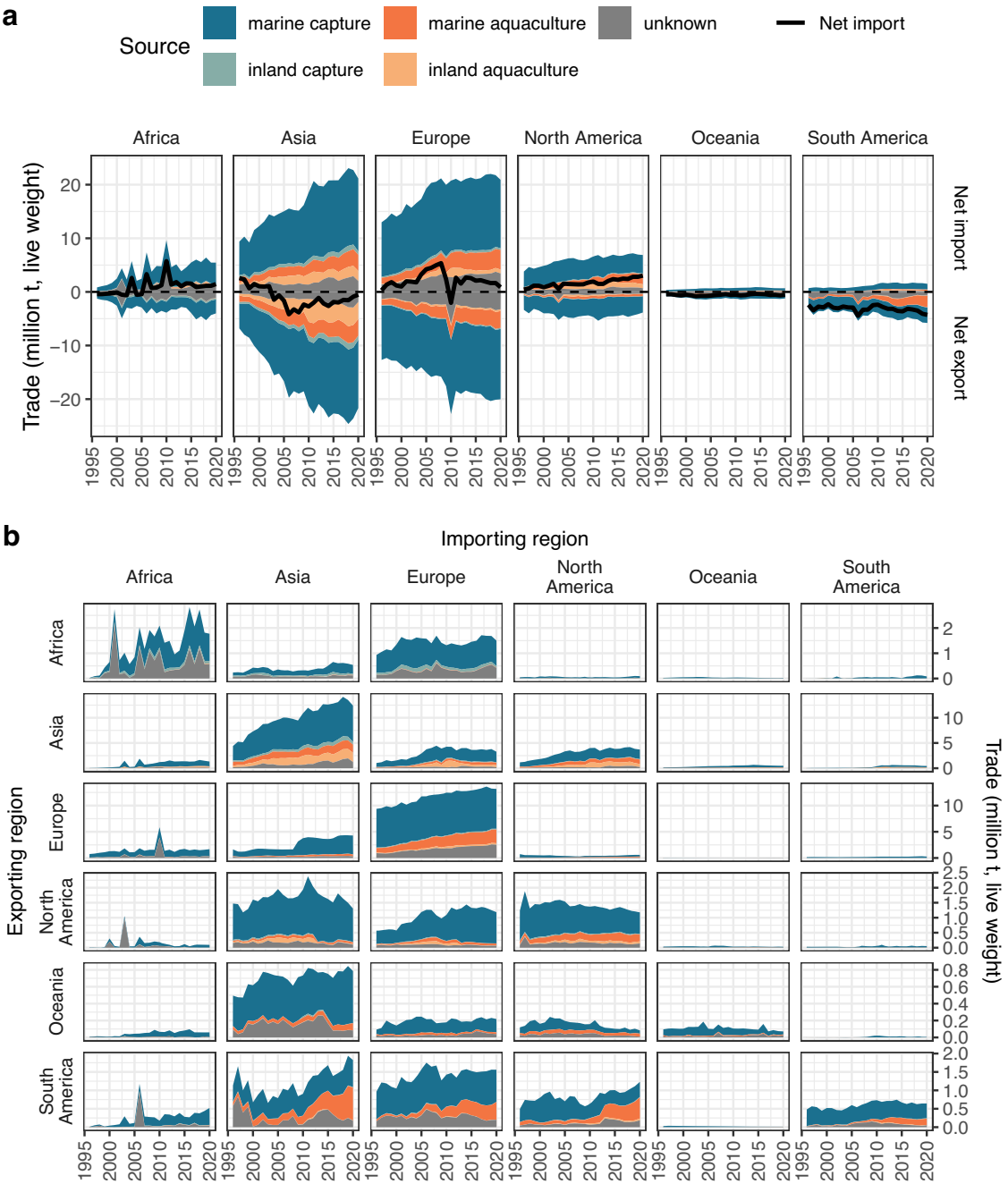


Fig. 3 | Regional trade flows by production source (habitat and method). a Total imports (positive) and exports (negative) colored by source, with net import trend in black. **b** Bilateral flows colored by production source with exporting region along the rows and importing region along the columns. Values represent a million tonnes in live weight equivalents. Note the *y*-axis scale differs for each row.

The increasingly globalized aquatic food system poses both challenges and opportunities for food security, sustainability, and resilience. Our work illuminates the evolution of farmed and wild aquatic food trade over the past 24 years, a period of rapid change for the sector. We show that trade patterns and trends differ across aquatic food groups, underscoring the value of species-resolved trade data. Marine capture remains the most highly globalized group, but aquaculture trade is growing faster. These trade patterns reflect major shifts within the industry, including the rise of foreign processing and the growth of aquaculture. Further, the ARTIS database presented here lays the foundation for answering pressing questions about the role of trade in meeting global food system goals.

Methods

To estimate the aquatic food species trade network, we compiled and aligned data on fishery and aquaculture production, live weight conversion factors, and bilateral global trade. The data span the globe and encompass decades of changes in country and species names and product forms. Over 4000 live weight conversion factors were compiled and matched to 2000+ farmed and wild capture aquatic species which in turn were matched to 900+ traded seafood product descriptions. Though we include nonfood (e.g., fish meal and ornamental trade) production and trade in the database, we exclude this from the analysis of aquatic food production and consumption. We also exclude mammals, reptiles, fowl, and seaweeds, along with

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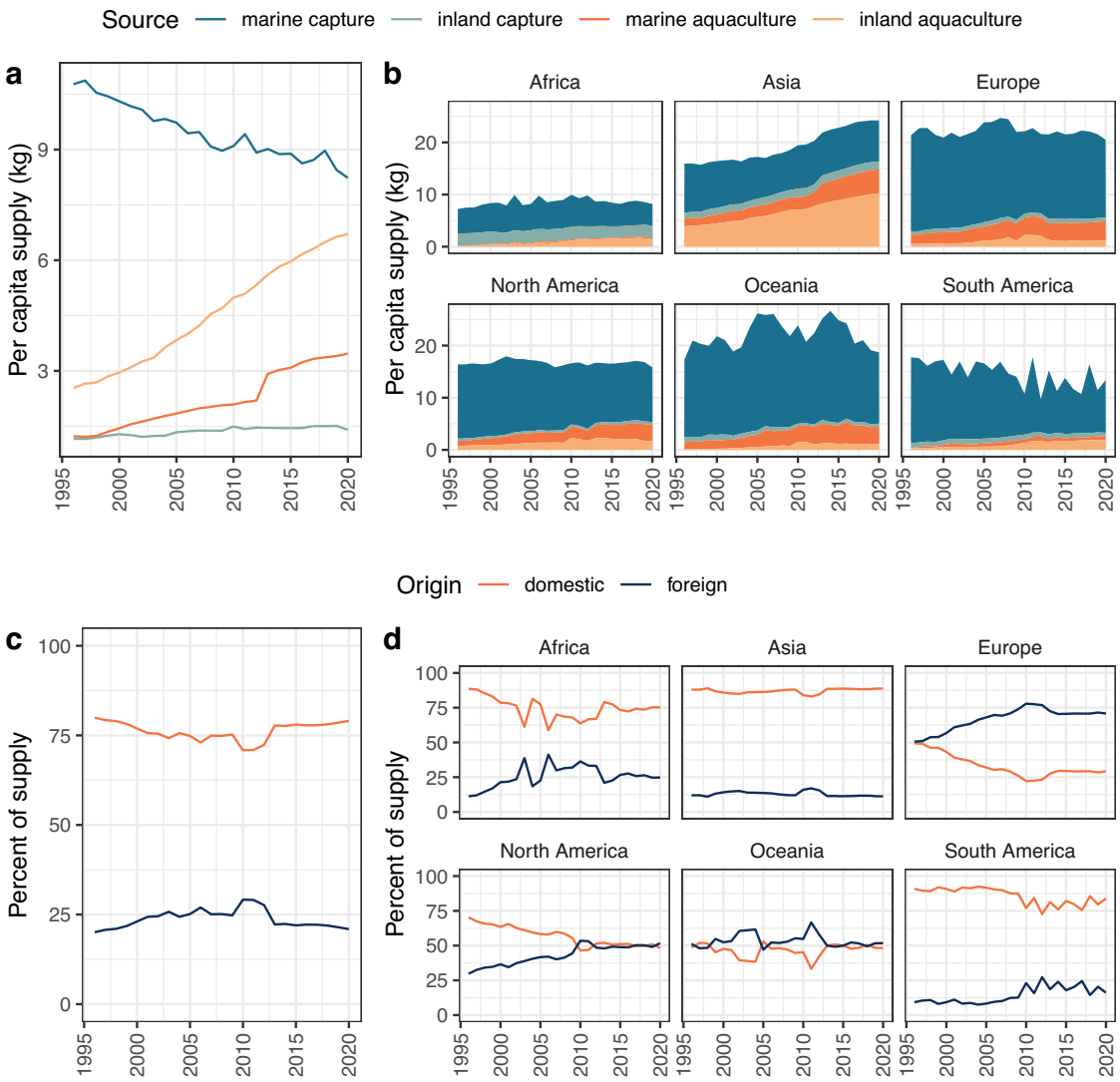


Fig. 4 | Aquatic food apparent consumption (supply) trends and regional patterns. a Global aquatic food apparent consumption by production source over time. **b** Regional aquatic food apparent consumption by production source over time. **c** Global aquatic food domestic vs foreign sourcing over time. **d** Regional aquatic food domestic vs foreign sourcing over time. Here, domestic refers to aquatic foods produced by the consuming country and foreign refers to aquatic foods produced by a different country.

co-products (e.g., caviar, shark fins, and fish meat) to avoid double counting, from the model and resulting database.

Species trade flow estimation occur in two steps. First, we take a mass balance approach, where each country's aquatic food exports must equal the domestic production, plus imports, minus domestic consumption, after accounting for processing losses. For each country, we estimate the proportion of production going into each possible commodity, the proportion of each imported commodity processed and exported, and the domestic consumption of each commodity. We then use these estimates with bilateral trade data to solve for the global species flows. This approach substantially improves upon previous efforts by estimating species-level trade, covering all production environments (marine and freshwater) and production methods (farmed and wild-caught), and including the processing and export of imported products.

ARTIS also differs from other available trade databases and models in important ways. Databases presenting reported bilateral trade, such as UN Comtrade, do not include the source country and are reported in terms of products which generally obscures the species and production method. Closely related or derived databases, such as BACI, thus have the same issues when trying to understand aquatic food origin and species (see below for more details on reported trade data). FAO also provides bilateral trade data on fishery products, with products at a greater disaggregation than what the 6-digit harmonized system (HS) offers, though this still does not provide sourcing or full species/production method information and the data is currently only available for 2019–2021. Multiregional input–output models do resolve flows to the source country and these methods could be adapted to derive a similar database once the species mix of domestically exported products is defined. However, existing input–output databases do not disaggregate aquatic foods at this resolution, generally representing "fisheries" and "food and beverages" as a single or in only a few sectors, as these models are generally constrained by the resolution of national input–output and supply and use tables. A notable exception is FABIO, which disaggregates food and agricultural flows into 130 agriculture, food, and forestry products, though seafood is represented as a single group[48].

Data

Production. Food and Agriculture Organization (FAO) national capture and aquaculture production data[5] serves as a key data input for disaggregating trade into improved species/species group resolution

in the ARTIS model. FAO compiles annual capture and aquaculture production data reported by around 240 countries, territories, or land areas from 1950 to 2020. Production data for around 550 farmed and 1600 wild capture species and species groups (e.g., *Bivalvia*) is reported in tonnes, expressed in the live weight equivalent. FAO production data consists primarily of official national statistics, with some verifiable supplemental information from academic reviews, consultant reports, and other specialist literature. Data reported by nations are checked by the FAO for consistency and questionable values are verified with the reporting offices. When countries fail to report production, FAO uses past values to estimate production. For the purposes of this analysis, we do not distinguish between nationally reported and FAO estimated values.

According to the Coordinating Working Party on Fishery Statistics, catch and landings should be assigned to the country of the flag flown by the fishing vessel irrespective of the location of the fishing. This means that production resulting from a country operating a fishing vessel in a foreign country's territory should be recorded in the national statistics of the foreign fishing vessel. However, if the vessel is chartered by a company based in the home country or the vessel is fishing for the country under a joint venture contract or similar agreement and the operation is integral to the economy of the host country, this does not apply. Consequently, our estimates of source country generally represent who harvested or caught the aquatic resource regardless of where it was produced (i.e., distant water fishing would generally be attributed to the flag state). In cases of exceptions related to select chartered foreign vessels, joint ventures, or other similar agreements, a catch by a foreign vessel but reported by the host country may not match trade reporting if the catch does not move through the customs boundary. These instances generate excess apparent consumption.

Bilateral trade data. We use the CEPII BACI world trade database, which is a reconciled version of the UN Comtrade database[49]. Trade data are reported to the UN by both importers and exporters following the HS codes. The HS trade code system organizes traded goods into a hierarchy, with the highest level represented by two-digit codes (e.g., Chapter 03 covers "Fish and Crustaceans, Molluscs and Other Aquatic Invertebrates"), which are broken down into 4-digit headings (e.g., heading 0301 covers "Live fish"), which are then subdivided into 6-digit subheadings (e.g., subheading 030111 covers "Live ornamental freshwater fish"). National statistics offices may further subdivide HS codes into 7- to 12-digit codes but since these are not standard across countries, the HS 6-digit codes are the most highly resolved trade codes available globally. HS codes are administered by the World Customs Organization, which updates the codes every five years. HS versions can be used from their introduction through the present, meaning that the HS 2002 version provides a time series of trade from 2002 to the present whereas the HS 2017 version only provides a time series back to 2017. Notably, HS version 2012 included major revisions to the HS codes relevant to fisheries and aquaculture products.

CEPII reconciles discrepancies in mirror trade records, which occur in around 35% of observations (for all traded commodities), by first removing transportation costs and using a weighting scheme based on each country's reporting reliability to average discrepancies in reported mirror flows. BACI data focuses on trade flows between individual countries since 1994 and therefore drops flows within some groups of countries (e.g., Belgium-Luxembourg) to ensure consistent geographies. The resulting data set covers trade for over 200 countries and 5,000 products. Further details on the BACI data set are available in ref. 49. While BACI resolves many data issues contained in the raw UN Comtrade database, it does not correct for all implausible trade flows, which can especially arise if one country misreports a value and the partner country does not report a value[50]. Further, there are instances where one country reports on trade that is optional, and the partner country does not. Here, we do not identify and re-estimate any values reported in BACI. Excessively large exports will generally result in high error terms, while high imports will result in high apparent consumption.

Trade statistics are managed by each territory and generally guided by the Kyoto Convention. For the purposes of trade data reporting, imports and exports represent all goods that add or subtract, respectively, from the stock of material resources within an economic territory, but not goods that merely pass through a country's economic territory. The economic territory generally coincides with the customs territory, which refers to the territory in which the country's customs laws apply. Goods that enter a country for processing are included in trade statistics. Goods that pass through a country "in transit," including those that are transshipped, are not recommended to be reported in trade statistics, though there are exceptions[51] and known instances where one country reports trade that is "in transit" but the partner does not, which creates discrepancies that are not corrected for within BACI. Fishery products from within the country, the country's waters, or obtained by a vessel of that country are considered goods wholly produced in that country. Catch by foreign vessels and catch by national vessels on the high seas landed in a country's ports are recorded as imports by the country the products are landed in and as exports by the foreign nation, where economically or environmentally significant. For further trade statistic guideline details, see ref. 51.

Live weight conversions. To satisfy a mass-balance problem, it is critical for the mass to be in the same units. However, trade data is generally reported in terms of the product weight (i.e., net weight), whereas production data is generally reported in the live weight equivalent. To convert product weight to the live weight equivalent, we include live weight conversion factors. We compiled live weight conversion factors from the European Market Observatory for Fisheries and Aquaculture Products (EUMOFA)[52] and national and international governmental reports. EUMOFA is updated annually and reports live weight conversion factors by CN-8 product code, the first 6 digits of which align with HS 6-digit codes, offering full coverage of traded aquatic food products. Therefore, we calculated the mean live weight conversion factor for each HS 6-digit code and drew most heavily from these values in the model. National and international reports that include live weight conversion factors often specify species-product form combinations, allowing us to better capture differences in product yields for different species. For national and international reports, we extracted the species name or categorized observations into taxa groups and identified the type of processing. To avoid double counting in the mass-balance model, we can only track the primary product. Consequently, we assign all co-products and byproducts a live weight conversion factor of zero, which effectively removes them from the model. Additionally, we assigned live animal trade a live weight conversion factor of 1 and assigned fishmeal an average value of 2.98[53].

Live weight conversion factors are an important source of uncertainty and error for studies conducting mass-balance analyses with trade data. There are several important limitations of live weight conversion factors: (1) only one value per product code can generally be used despite the fact that some product codes include forms with vastly different live weight conversion factors (e.g., bivalve codes that include both shell on and shell-off products); (2) the species is often not specified, and even when it is, live weight conversion factors vary for individuals of different sizes; and (3) source of the data and information on the country and time point for values are often omitted, which represents a problem because there is known geographical and temporal variation in live weight conversion factors due to differences in processing technology.

All conversion factors were reported as live weight-to-product weight ratios. These conversion factors were mapped onto possible

species-to-commodity or commodity-to-commodity conversions, described below. For commodity-to-commodity conversions, we estimate the conversion factors (i.e., processing loss rate) as the additional mass lost when converting from the live weight to the original product form relative to converting from the live weight to the processed product form. This can be calculated as the live weight conversion factor for the original product form divided by the live weight factor for the processed product form. We assume that mass cannot be gained through processing and therefore impose a maximum value of one to this ratio.

Seafood production and commodity conversion. For each country-year-HS version combination, we estimate the proportion of each species going into each commodity and the proportion of each imported commodity processed into each other commodity. Each species can only be converted into a subset of the commodities. For example, Atlantic salmon, *S. salar*, can be converted into whole frozen salmon or frozen salmon filets, but cannot be converted to a frozen tilapia filet. Similarly, each commodity can only be converted to a subset of other commodities through processing. For example, whole frozen salmon can be processed into frozen salmon filets, but not vice versa and neither salmon commodity can be converted to a tilapia commodity through processing. Defining possible conversions restricts the solution space to realistic results and improves estimation by reducing the number of unknowns. We describe this assignment process in detail below.

Taxonomic group to commodity assignment. A single commodity or product could include multiple species (e.g., a salmon fillet code could include multiple species of salmon) and a single species could be processed into multiple product forms (e.g., salmon fillets, whole salmon, or canned salmon). This creates a many-to-many matching problem. To match species to their appropriate codes, we extracted taxonomic information from each HS code and drew upon the taxonomic information for each species name appearing in the FAO production data. Additionally, we drew upon the HS system hierarchy to create matches for species "not elsewhere considered" within a given heading level.

To ensure species groups were matched to the correct 4-digit heading, we divided all species into finfish, crustaceans, mollusks, and other aquatic invertebrates. Within each group, we matched species to 6-digit codes according to the following:

1. Explicit taxa match—species or higher taxonomic information related to the species is included in the code description.
2. NEC match—species that fall within a 4-digit heading but are not included as an explicit taxa match within that heading are matched to the associated "not elsewhere considered" (NEC) code.
3. NEC by taxa match—the same matching logic as an NEC match, but restricted to a specified taxonomic category (e.g., Salmonidae, NEC).
4. Broad commodity match—if no specific taxonomic information is included, then the match occurs based on the broad taxonomic groups.
5. Aquarium trade match—species that have been identified as part of the global aquarium/ornamental trade[54].
6. Fishmeal—assigned to fishmeal codes if at least 1% of production goes to fishmeal production globally during the study period based on the end-use designation from Sea Around Us production data[55]. Although an estimated 27% of fishmeal is derived from processing by-products[3], the species, geographical, and temporal variation in that estimate is currently unknown. Consequently, fishmeal is currently treated as sourced from whole fish reduction. This does not affect the total trade or trade patterns of fishmeal but does result in an overestimate of the proportion of production going to fishmeal in cases where by-products are used.

After all species are matched to the appropriate HS codes, we use the list of species to define codes as inland, marine, diadromous, or mixed. Higher-order taxonomic groups are then only matched with HS codes that include their habitat. For example, the production of inland *Actinopterygii* is matched with codes that include inland species that fall within *Actinopterygii*, but not with exclusively marine codes, even if they contain species that fall within *Actinopterygii*.

Commodity to commodity processing assignment. As with the species-to-commodity assignment, the commodity-to-commodity assignment is a many-to-many data problem. Here, one commodity can be processed into multiple other commodities (i.e., frozen salmon can be processed into salmon filets or canned salmon), which also means one commodity could have come from multiple other commodities. To create these assignments, we established rules for which product transformations are technically possible. First, a product cannot transfer outside of its broad commodity group (e.g., fish, crustaceans, mollusks, aquatic invertebrates). Second, where a more refined species or species group was given (e.g., tunas, salmons, etc.) a product cannot be transformed outside that group. Third, products are classified in terms of their state (e.g., alive, fresh, frozen, etc.) and presentation (e.g., whole, fileted, salted/dried/preserved meats, reductions such as fish meal and fish oil, etc.) and cannot be converted into less processed forms (e.g., frozen salmon filets cannot turn into a frozen whole salmon). Fourth, specific commodities (i.e., mentioning specific species) and NEC commodities can become broad commodities (where appropriate), however, broad commodities cannot become more specific or NEC commodities.

Country standardization and regions. The FAO production and BACI trade datasets do not share the same set of countries and territories. For the production and trade data to balance, it is important for the set of territories falling under a given name to align across the datasets. To avoid instances where, for example, production is reported under a territory, but trade is reported under the sovereign nation, we generally group all territories with the sovereign nation. As countries gain independence, they are added as a trade partner in the database. Due to this country standardization circular flows may occur when a sovereign nation trades with their territory. These circular flows are filtered out of the standardized BACI trade flows (i.e., internal trade is not included).

Network estimation

Estimating species bilateral trade flows occurs in two steps: first, solving the national production-trade mass balance, and second, converting reported commodity trade flow estimates to species trade flow estimates based on the estimated species mix going into each domestic and foreign exported commodity.

National mass-balance. We start with the fact that exports must equal production and imports, minus consumption. Since exports are reported as commodities, we solve this mass balance problem in terms of commodities. Production data are reported for each species, so we estimate the elements of a matrix that represents the proportion of production going into each commodity. Since an imported commodity can be processed and exported as a different commodity, we also estimate the proportion of each import being converted into a different commodity. Then for a given country,

$$e = V_1 \circ X \cdot p + V_2 \circ W \cdot g - c + \epsilon \quad (1)$$

If n is the number of species and m is the number of commodities, then: V_1 is a sparse $(m \times n)$ matrix with product conversion factors corresponding to the unknowns in X; X is a sparse $(m \times n)$ matrix of the

proportion of each species in each commodity; p is a vector of domestic species production ($n \times 1$); V_2 is a sparse ($m \times m$) matrix with product conversion factors corresponding to the entries of W; W is a ($m \times m$) matrix of the processed imported commodities; g be a vector of imports ($m \times 1$), c is a vector of domestic consumption ($m \times 1$), and ϵ is a vector of error terms ($m \times 1$).

We compiled reported values for V_1, V_2, e, p, and g, and estimated the entries of X, W, c, and ϵ. We first converted this problem to a system of linear equations. Using the property that $\mathrm{vec}(ABC) = (C^T \otimes A)\mathrm{vec}(B)$, we can create $A_b = (y^T \otimes D_m)D_V$, where D_m is a diagonal matrix of ones, with dimension m and D_V is a diagonal matrix with the elements of $\mathrm{vec}(V)$. The vector of unknowns is then $x_b = \mathrm{vec}(Z)$. We then solve this system of equations with a quadratic optimization solver such that the mass balance equalities are satisfied, trade codes with higher species resolution in X are prioritized, and the elements of X, W, and c are otherwise relatively even (i.e., we assume an even distribution of production among commodities unless the data suggests otherwise), that ϵ is as small as possible (i.e., minimize the error), and all unknowns are greater than or equal to zero.

Positive error terms represent situations where reported production and imports cannot explain exports. This can occur due to under- or un-reported production or imports, over-reporting of exports, errors in the live weight conversion factors, or inconsistencies in the year production and trade are attributed to.

We solve the mass-balance problem for each country-year-HS version combination using the Python package "solve_qp." The estimated species mixes in national production (X), processing of imports (W) and the error term (ϵ) are passed to the next stage of the analysis.

Converting the product trade network to a species trade network. First, we compute the mix of species going into each trade code for each country's domestic exports. To do this, we reweight X so it represents the proportion of each species in each code rather than the proportion of production of a species going into each product. Each country's estimated X matrix is multiplied by p to get the mass of each species in each commodity. The total mass of each commodity is found by summing all the species volume grouped by commodity and the proportion of each species within a commodity is then calculated by dividing all volumes by their respective commodity mass totals.

Each country's exports can be sourced from domestic production, imported products that are subsequently exported, with or without processing (i.e., foreign exports), or from an unknown source (i.e., error exports). Since the mix of these sources cannot be derived from the mass balance equation alone, we calculate a range for sourcing following[33]. We calculate the maximum possible domestic exports by taking the minimum between the domestic production and total exports. Similarly, we calculated the maximum volume of exports sourced from imports, by taking the minimum between each product's imports (accounting for processing estimated by W) and exports. The minimum domestic exports are calculated as the minimum between production and the difference in exports and the maximum calculated foreign exports, with the remainder as error exports (minimum foreign exports are calculated in an analogous way). The above results represent midpoint estimates.

$$e_{domestic,\max} = \min(V_1 \circ X \cdot p, e) \tag{2}$$

$$e_{foreign,\max} = \min(V_2 \circ W \cdot g, e) \tag{3}$$

$$e_{domestic,\min} = \min(p, e - e_{foreign,\max}) \tag{4}$$

$$e_{foreign,\min} = \min(g, e - e_{domestic,\max}) \tag{5}$$

$$e_{domestic,mid} = \frac{e_{domestic,\max} + e_{domestic,\min}}{2} \tag{6}$$

$$e_{foreign,mid} = \frac{e_{foreign,\max} + e_{foreign,\min}}{2} \tag{7}$$

For these three estimates (maximum, minimum, and midpoint) we calculate the domestic and foreign weights by dividing domestic export values and foreign export values by total export. We then distribute each country's exports into domestic, foreign, and error exports by multiplying exports by domestic, foreign, and error proportions (Fig. S8). For each export source, we apply a different species mix to each HS code based on the estimated source country. For domestic exports, we use the exporting country's estimated X matrix (Fig. S9). For error exports, the geographical origin is unknown and may arise from unreported production, so we cannot meaningfully assign a species mix to the code. Consequently, we identify the lowest taxonomic resolution common to all species within the code and assign that name to the trade flow.

For foreign exports, we trace the origins back in the supply chain a maximum of three steps (i.e., producer to the intermediate exporter to final exporter to final importer), with any remaining foreign export or flows less than 1 tonne left as "unknown" source (Supplementary Fig. 8). The small flows left unresolved comprise around 1% of total trade (Supplementary Fig. 8). To link an export of foreign origin to its source country, we use a reweighted version of W to estimate the original imported product codes and connect those to their source country, using a proportional breakdown of each country's imports of that code. Foreign exports of one country that originated from foreign exports of another country are isolated and undergo the process above to identify the source country. The species mix for foreign trade flows is based on either the source country's estimated X matrix or the method described above for error exports (Supplementary Fig. 9).

Network post-estimation processing

Once the species trade flow network is built, we remove all volumes traded below 0.1 tonnes, as the multiplication by small proportions generates overly specific, and likely unrealistic, small flows.

Next, to generate a complete time series, we need to compile estimates from across the HS versions. All HS versions are reported since they have been created, for example, HS96 reports trade from 1996 until the present. However, the more recent HS versions generally include more specific trade codes and therefore are preferred over older versions. It takes a few years before an HS version is fully adopted, resulting in lower total trade volumes for the first few years an HS version is available compared to the previous HS versions (Supplementary Fig. 7). To provide the most accurate representation of trade, we create a continuous time series by adopting the most recent HS version available after its total trade has met up with the total trade reported under previous HS versions. This results in HS96 being used for 1996–2004, HS02 for 2004–2009, HS07 for 2010–2012 and HS12 for 2013–2020.

To check the reasonability of estimated trade flows, we first confirmed that all trade flows sum to the original BACI trade flows when grouped by exporter, importer, year, and HS code and expressed as product weight. Note that some flows are slightly lower due to the 0.1-tonne threshold (maximum difference of 72 tonnes representing a percent difference of 0.19%). Second, we confirmed that the estimates from the mass balance problem satisfy the problem constraints. Third, we checked that domestic exports of species in live weight equivalent do not exceed the production of that species. Fourth, we confirmed that exports of foreign sources do not exceed imports of that species. Only 1.4% of cases across all years showed a country's foreign export of a species exceeded the total import of that species.

The model was implemented in R and all packages used are listed in Supplementary Table 1.

Analysis

Calculation of apparent consumption (supply). The species trade network and FAO production data were used to calculate national apparent consumption by scientific name, habitat, and method for each year.

First, domestic production of products is estimated by multiplying production data by the corresponding estimated X matrix. We then calculate domestic consumption by subtracting domestic exports by HS code from domestic production of products. Domestic consumption by species is then derived based on the volume of domestic consumption and the estimated species composition for the associated code.

Foreign consumption represents the quantity of product imported that was consumed in the country (i.e., not subsequently exported). To calculate foreign consumption, we subtract foreign exports from the quantity of processed imports. Processed imports represent the quantity of each product, by HS code, available after accounting for processing, by multiplying the appropriate estimated W by the import vector i. We convert processed imports to live weight by multiplying by the live weight conversion factor. To distinguish human consumable products, we filter out all products not destined for direct human consumption.

Finally, we disaggregated foreign consumption of processed HS products to species. We assume the species and trade sourcing distribution of foreign consumption of a given code is proportional to the species distribution of the original imported HS products from which a final code was sourced based on the estimated species trade network. We therefore disaggregate foreign consumption by multiplying foreign consumption by the trade flow proportions of imports across all trade partners and species information.

Since apparent consumption is based on a disappearance model, estimated values are subject to multiple sources of error. Due to discrepancies in production and trade reporting for select countries (e.g., as arises with joint ventures), a few countries had unrealistically large estimated per capita consumption. For country-specific consumption estimates, we capped total per capita consumption to 100 kg, as this is slightly above the upper estimate FAOSTAT[25]. We then adjusted the supply by export partners, scientific name, production method, habitat, and source country for those countries proportionally. A second factor that influences estimated apparent consumption relates to the approach for removing non-human consumable products, particularly the domestic production and use of fishmeal. While we base our estimates on the species mix entering HS code 230120, this is estimated based on exports and domestic use patterns could diverge, leading to errors in the volume removed. Additionally, since many species can enter code 230120 and there is limited empirical data to inform volumes of species entering fishmeal by country, there is greater uncertainty in the exact species mix, and therefore greater uncertainty in the species volumes to exclude from direct human consumption calculations.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All input data, key intermediate data files, and data files required to reproduce these figures/results are archived in Zenodo (https://doi.org/10.5281/zenodo.10034319). All input data for running the full ARTIS model are archived on KNB (https://doi.org/10.5063/F1862DXT) and the full ARTIS database is archived on KNB (https://doi.org/10.5063/F1CZ35N7). Source data are provided with this paper.

Code availability

The code to reproduce all results and figures is available at https://github.com/jagephart/ms-seafood-globalization and an archived version is available on Zenodo (https://doi.org/10.5281/zenodo.10034319). The code for the model underlying the ARTIS database is available at https://github.com/Seafood-Globalization-Lab/artis-model/releases/tag/v1.0 and an archived version is available on KNB (https://doi.org/10.5063/F1CZ35N7).

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Author contributions

The ARTIS database was conceptualized by J.A.G. and analysis of the database was conceptualized by J.A.G., C.D.G., R.L.N., and M.L.P. The methodology, software, formal analysis, and data curation were conducted by J.A.G., R.A.B., and K.G. The contribution to the visualizations presented by J.A.G., R.A.B., and A.G. The original draft was prepared by J.A.G., R.A.B., and K.G. The contribution to the review and editing by J.A.G., R.A.B., K.G., A.G., C.D.G., R.L.N., K.L.N., M.L.P., and M.T. The funding for this work was acquired by J.A.G., A.G., and C.D.G.

Competing interests

The Authors declare the following competing interests: J.A.G., C.D.G., and R.L.N. are members of the Oceana Science Advisor Board. The remaining authors declare no competing interests.

Additional information

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